

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

Anonymous authors
Anonymous submission

ABSTRACT

Diffusion probabilistic models (DPMs) have demonstrated remarkable progress in generative tasks, such as image and video synthesis. However, they still often produce hallucinated samples (*hallucinations*) that conflict with real-world knowledge, such as generating an implausible duplicate cup floating beside another cup. Despite their prevalence, the lack of feasible methodologies for systematically quantifying such hallucinations hinders progress in addressing this challenge and obscures potential pathways for designing next-generation generative models under factual constraints. In this work, we bridge this gap by focusing on a specific form of hallucination, which we term **counting hallucination**, referring to the generation of an incorrect number of instances or structured objects, such as a hand image with six fingers, despite such patterns being absent from the training data. To this end, we construct a dataset suite **CountHalluSet**, with well-defined counting criteria, comprising ToyShape, SimObject, and RealHand, on which our counting models reach over 99% counting accuracy and thus make the reported hallucination rates directly measurable rather than estimated. Using these datasets, we develop a standardized evaluation protocol for quantifying counting hallucinations, and systematically examine how different sampling conditions in DPMs, including solver type, ODE solver order, sampling steps, and initial noise, affect counting hallucination levels. The resulting measurements reveal a clear and repeatable pattern: strategies that improve perceptual quality do not reduce counting hallucinations, and on RealHand raising the sampling budget from 25 to 100 steps leaves the counting hallucination rate essentially flat or slightly higher, whereas the same change lowers it on the simpler synthetic datasets. Furthermore, our correlation analysis quantifies this gap directly: perceptual metrics such as Fréchet Inception Distance (FID) track non-counting failures strongly (all coefficients at least 0.88) yet correlate weakly and inconsistently with counting hallucinations, demonstrating that widely-used perceptual metrics fail to capture counting hallucinations consistently.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), commonly referred to as diffusion model, have achieved considerable success in generative modeling, producing high-quality samples across various tasks such as text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). Despite these impressive advancements, diffusion models still struggle to faithfully capturing factual properties from training data, often generating outputs that violate consistency with real-world knowledge. A notable example is the generation of objects with an incorrect number of spatially intricate components, such as extra human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023).

Recent studies have increasingly focused on identifying and analyzing common failure modes in text-guided diffusion models. For instance, Borji (2023) systematically concluded several qualitative shortcomings in text-to-image models. Additionally, Liu et al. (2024) proposed an adversarial search method to uncover undesirable behaviors and failure cases in text-to-image generation, while Wu et al. (2023) employed an iterative approach to discover and mitigate spurious correlations.

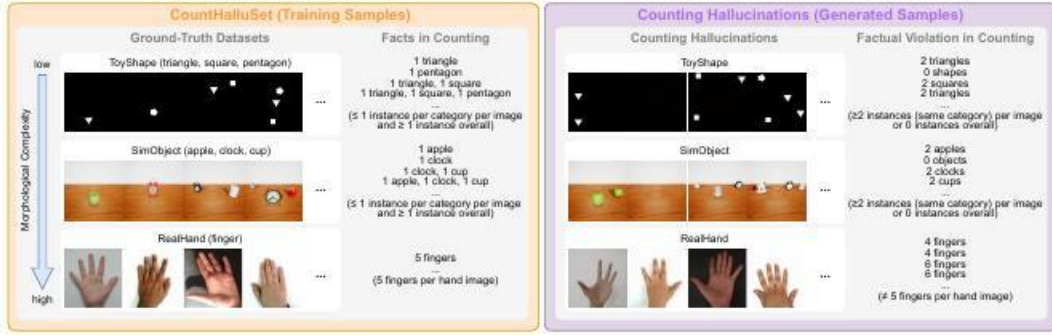


Figure 1: Example comparison between the proposed **CountHalluSet** used for training (left) and the corresponding **counting hallucinations** generated by the model (right). Each dataset contains different objects with specific counting criteria. For example, in ToyShape and SimObject, each image contains at most one instance per category and at least one instance overall, whereas in RealHand, each image contains exactly five fingers. Counting hallucinations refer to generated images that violate the dataset’s counting rules, such as two apples in an image or an image containing no objects but a normal background, even though these patterns never appear in the SimObject dataset.

More recently, *hallucination*, a specific failure mode in diffusion models, has garnered attention in the research community (Aithal et al., 2024). Unlike previously discussed text-prompt-based failure modes, which can be evaluated by measuring the alignment between the input conditions and outputs (e.g., CLIP score (Hessel et al., 2021) and VQA score (Huang et al., 2025; Zarei et al., 2024)), or more obvious failure modes such as severe distortions, blur, and artifacts, hallucinated samples (*hallucinations*) often appear visually plausible, and emerge independently of the conditioning signal, manifesting as subtle violations of factual consistency with respect to the training data, such as incorrect object counts, geometries that appear plausible but are physically impossible, or other breaches of physical laws. These fine-grained distinctions between valid samples and hallucinations present a critical challenge for current AI systems, which often fail to detect them reliably. As a concrete indication of how hard reliable counting is, even state-of-the-art large vision-language models like GPT-4o still struggle with a task as basic as accurately counting the fingers in a clear hand image, which is precisely why we build dedicated, human-validated counting models rather than relying on off-the-shelf detectors. To this end, establishing a standardized and well-defined protocol for reliably quantifying hallucinated samples is essential, as it provides a foundation for outlining pathways toward designing next-generation generative models with strict factual consistency, an ability that is critical for their role as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Among the various common types of hallucinated elements, such as counts, colors, spatial relationship, geometry, and physical realism, counting errors are particularly suitable for systematic study because they are not only easy to identify but also, crucially, straightforward to quantify: an integer count either matches the predefined criterion or it does not, yielding an unambiguous per-image label and a well-defined rate that can be compared cleanly across conditions. This measurability motivates us to begin by defining a specific form of hallucination termed **counting hallucination**. *In this setting, the number of objects in training data is explicitly predetermined. Consequently, a generated image is deemed hallucinated whenever its object counts deviate from these predefined values.* For example, an image depicting a hand with six fingers is classified as a counting hallucination, since such a pattern never occurs in the RealHand dataset, as shown in the last row of Fig. 1. To facilitate the study of such phenomena, we construct a dataset suite, CountHalluSet, consisting of three datasets that span a spectrum of morphological complexity for the countable objects: ToyShape (*triangle, square, pentagon*), SimObject (*mug, apple, clock*), and RealHand (*finger*). Figure 1 illustrates representative examples from these datasets alongside corresponding counting hallucinations.

In our quantification protocol for counting hallucinations, a critical step is the accurate identification and quantification of objects. To this end, we employ counting models tailored to each dataset, each validated against human annotations to exceed 99% counting accuracy so that the reported rates reflect genuine model behavior rather than measurement noise. Leveraging these datasets and counting models, we quantify how various denoising conditions (e.g., sampling steps, initial noise, the order of ODE solvers, and solver type) within diffusion models affect counting hallucination

rates, holding every other factor fixed so that each comparison isolates a single variable. Besides, we conduct correlation analysis between counting hallucination rate and perceptual metrics, including Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021), contrasting how tightly each metric tracks counting hallucinations versus non-counting failures.

Our study yields three findings, each backed by consistent quantitative evidence across solvers, step budgets, and datasets:

- Commonly applied numerical strategies fail to mitigate counting hallucinations on human hands: increasing sampling steps from 25 to 100 does *not* lower the counting hallucination rate on RealHand (it stays near 13–15% and even rises for the first-order solver), even though the identical change lowers the rate on the simpler synthetic ToyShape (2.43% $\rightarrow$ 1.56%) and SimObject (9.27% $\rightarrow$ 8.04%) datasets.
- Perceptual metrics separate the two failure types sharply: the counting hallucination rate correlates only weakly and inconsistently with FID and MUSIQ (correlations flip sign across datasets), whereas the non-counting failure rate correlates strongly and consistently with the same metrics (all coefficients at least 0.88).
- Building directly on these findings, we propose joint-diffusion models (JDM), a training paradigm that consistently lowers *both* counting hallucinations and non-counting failures on RealHand across every solver and step setting, reducing the non-counting failure rate by up to roughly $6\times$ relative to the vanilla LDM baseline.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

Inspired by the growing attention to hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), the study of hallucinations in unconditional diffusion models has also garnered increasing interest. In particular, Aithal et al. (2024) proposed a qualitative hypothesis suggesting that hallucinated samples may arise from data interpolating between distinct modes. Despite this insight into potential causes, the analysis remains qualitative; a systematic, number-driven quantification of hallucinations in diffusion models, one that reports comparable rates across controlled conditions, is still lacking.

This gap motivates us to facilitate the accurate identification and quantification of hallucinations in diffusion-generated outputs. By reducing hallucination to a concrete, reproducible rate, such an investigation can deepen our understanding of the undesirable behaviors exhibited by these models and quantify precisely how the magnitude of denoising errors, such as variations in sampling steps or the order of ODE solvers, shifts the severity of hallucinations.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

The Visual Question Answering (VQA) score (Huang et al., 2025; Zarei et al., 2024) and the CLIP score (Hessel et al., 2021) are two widely used metrics for evaluating the performance of text-guided generated images. However, such evaluation methods that rely on language-based models are inherently limited when applied to assess hallucinated images. One line of evidence is the statistical lower bound on the propensity of pretrained language models to hallucinate certain fact types, which constrains the upper bound of detection accuracy based solely on language models (Kalai & Vempala, 2024). Another indication is the persistent performance gap between human annotators and LLMs in classifying hallucinated textual content (Lin et al., 2022; Li et al., 2023).

In addition, several widely used quantitative evaluation metrics, such as FID (Heusel et al., 2017), IS (Salimans et al., 2016), and Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021) are commonly employed in image generation tasks. These metrics rely on the feature embeddings extracted from a pretrained Inception-v3 model (Szegedy et al., 2016). However, *whether the numerical values of these metrics actually track the severity of hallucinations in generated datasets* remains an open question. In this work, we answer it quantitatively from the perspective of counting hallucinations, reporting side-by-side correlations that show these perceptual scores align with non-counting failures but not with counting hallucinations.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given the observed data $x_0 \sim q_0(x_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) introduce a *forward process* (a.k.a. *diffusion process*) that gradually perturbs the data by adding Gaussian noise over T discrete timesteps. At each timestep t , the marginal distribution of the noisy sample x_t conditioned on the original data x_0 follows:

$$q_{0:t}(x_t|x_0) = \mathcal{N}(x_t | \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_i\}_{i=1}^T$ is a pre-defined noise schedule dependent on t , commonly chosen to be linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021) schedules. The factor $\bar{\alpha}_t$ controls the signal-to-noise ratio at step t : $\bar{\alpha}_0 = 1$ (no noise) and $\bar{\alpha}_T \approx 0$ (nearly pure noise). An equivalent formulation of Eq. (1) is:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + (1 - \bar{\alpha}_t)\epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

As T becomes sufficiently large, the marginal distribution $q_{0:T}(x_T|x_0)$ approaches a standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. In widely-used DPMs (Ho et al., 2020; Nichol & Dhariwal, 2021), T is typically set to 1000, the same value we adopt throughout our experiments to keep all reported comparisons on a common footing.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (a.k.a., *denoising process*) of DPMs aims to iteratively recover the observed data based on the diffused data x_T . Specifically, DPMs attempt to predict the noise ϵ from the data x_t by using a neural network $\epsilon_\theta(x_t, t)$, known as *noise prediction model*. The parameter θ is optimized by minimizing the following objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, x_0, \epsilon} [\pi(t) \|\epsilon_\theta(x_t, t) - \epsilon\|_2^2], \quad (3)$$

where $\pi(t) > 0$ is a weighting function.

Ancestral Sampling. Given noisy data x_T , samples from the original distribution $q_0(x_0)$ can be obtained by reversing the diffusion process with the same number of steps, commonly referred to as *ancestral sampling*:

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1} | \frac{1}{\sqrt{1 - \beta_t}}(x_t + \beta_t \nabla_{x_t} \log q_t(x_t)), \beta_t \mathbf{I}), \quad (4)$$

where $\nabla_{x_t} \log q_t(x_t)$ is the only unknown term, referred as *score function* (Song et al., 2022).

In practice, DPMs approximate the scaled score function $-\sqrt{(1 - \bar{\alpha}_t)} \nabla_{x_t} \log q_t(x_t)$ via the noise prediction model. Therefore, an equivalent formulation of Eq. (3) is:

$$x_{t-1} = \mu_\theta(x_t, t) + \sqrt{\beta_t}\epsilon, \quad \mu_\theta(x_t, t) = \frac{1}{\sqrt{1 - \beta_t}}x_t - \frac{\beta_t}{\sqrt{1 - \beta_t}} \frac{1}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. The ancestral sampling in Eq. (4) contains a stochastic term (Kloeden et al., 1992), while there is an associated deterministic process, which can be formulated as an ordinary differential equation (ODE) (a.k.a. *probability flow ODE* (Song et al., 2022)):

$$\frac{dx(\tau)}{d\tau} = f(\tau)x(\tau) + h(x(\tau), \tau), \quad f(\tau) = -\frac{1}{2}\beta(\tau), \quad h(x(\tau), \tau) = -\frac{1}{2}\beta(\tau)\nabla_{x(\tau)} \log q_\tau(x(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $x(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h(x(\tau), \tau)$ and starts from an ideal diffused data $x(T)$ drawn from the true diffused distribution $q_T(x(T))$. Set the discrete time steps as $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, where $\tau_N = 0$. Applying the variation-of-constants formula for the exact solver over one step:

$$x(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k)x(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u)h(x(u), u)du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v)dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$. $\Delta t > 0$ denotes the step size, allowing ODE-based solvers to accelerate the sampling process with larger step sizes compared to ancestral sampling. The total number of sampling steps is given by $T/\Delta t$. Since the linear part $\mathcal{G}(t_2, t_1)$ can be computed in closed form, most accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) devote their numerical efforts, such as higher solver orders and more sampling steps, to accurately approximating the nonlinear integral term involving h . These two knobs, solver order and step count, are exactly the numerical strategies whose effect on counting hallucinations we quantify in Sec. 5.2. More theoretical analysis can be found in Appendix A.1 to A.3.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

In this section, we describe the methodology and datasets that turn counting hallucination into a precisely measurable quantity. We first define counting criteria and the associated counting hallucinations, then introduce the datasets in the CountHalluSet suite. For each dataset, we specify its counting criteria and corresponding counting hallucinations, and outline the standardized protocol used for their quantification, so that every rate reported later in the paper is comparable across datasets and sampling conditions.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

In this section, we provide a general definition of counting criteria within training datasets and define counting hallucinations as deviations from these predefined constraints.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ be a training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, the number of objects of category c in $\mathbf{x}$, $N_c(\mathbf{x})$, are given by a predefined set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$), i.e., $N_c(\mathbf{x}) \in \mathcal{S}_c$. We further require that a valid reference sample contains at least one object, i.e. $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C is the set of considered categories.

Counting hallucinations as deviations from the counting criteria. Given a generated sample $\hat{\mathbf{x}}$, we say $\hat{\mathbf{x}}$ exhibits a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ if both of the following conditions hold:

1. **Sufficient visual quality for counting.** The image $\hat{\mathbf{x}}$ should exhibit sufficient visual quality such that the background and any present objects are clearly discernible. This ensures that counting hallucinations can be distinguished from other prominent failure modes, such as severe degradations. We define a binary indicator $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, referred to as the *counting-ready indicator*, where 1 denotes that the image quality is sufficient for counting, and 0 indicates otherwise. In practice this indicator can be determined via human annotation, by a classifier calibrated against human annotations, or by other automated criteria.
2. **Violation of counting facts.** A violation of counting facts occurs if there exists a category c such that $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or if the image contains no objects (i.e., $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$) as the dataset requires at least one object per sample. In practice, we use a counting model to predict object counts in generated images as evidence for such violations.

A counting hallucination (CH) is identified if its quality is sufficient for counting ($\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$) and the predicted counts deviate from the counting criteria ($\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

The CountHalluSet suite consists of three datasets for studying counting hallucinations: ToyShape, SimObject, and RealHand. Examples from each dataset and corresponding counting hallucinations are shown in Fig. A1 and A2. Additional statistical summaries are provided in Appendix A.4.

4.2.1 TOYSHAPE

Dataset description. The ToyShape dataset comprises 30,000 images built from three geometric shapes: *triangle*, *square*, *pentagon*. All shapes are white with an identical area of 120 pixels and never overlap, so the only factor distinguishing a valid image from a counting hallucination is the count itself, making ToyShape the cleanest testbed in the suite. The counting criteria specify that each image contains at most one instance of each shape category and at least one shape (in total), as shown in the top-left of Fig. 1. Formally, we have $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in S_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Counting hallucinations. According the counting criteria defined above, any image generated by the model trained on ToyShape that contains two or more shapes of the same category is considered a counting hallucination, such as an image containing two pentagons in the top-right region of Fig. 1. Besides, empty images (i.e., containing only the black background without any shapes) are also classified as counting hallucinations. Examples can be found in Appendix A.5.

Counting model used in ToyShape. We utilize a counting model to identify counting errors within images, without using a counting-ready indicator, as the dataset’s simplicity makes severely degraded images rare. In specific, we fine-tune a ResNet-50 (He et al., 2016) using over 400k ToyShape data, with each category appearing 0–3 times per sample (i.e., $S_c = \{0, 1, 2, 3\}$), aiming to capture a wide range of plausible scenarios, including counting-valid samples and counting hallucinations, enabling accurate quantification. On generated data the resulting counting model attains over 99.9% counting accuracy, so the ToyShape hallucination rates reported below are effectively exact.

Quantification of counting hallucinations. Given a diffusion model trained on the ToyShape dataset, we generate a set of samples equal in size to the training dataset (30,000 images), matching the training count so that generated and reference distributions are compared at the same scale. We then employ the counting model to predict the number of objects in each generated image. Lastly, images that violate the counting criteria are identified as counting hallucinations, and the counting hallucination rate is simply their fraction of the 30,000 samples, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. The SimObject dataset includes 30,000 rendered images of everyday objects, matching ToyShape in scale so that hallucination rates are compared at equal sample size. It spans three object categories, *mug*, *apple*, and *clock*, each with 10 distinct intra-class variations, placing it at an intermediate level of morphological complexity between ToyShape and RealHand. All objects are randomly placed on a wooden tabletop under fixed lighting conditions, with minimal mutual occlusion if any, as demonstrated in the middle of the left side of Fig. 1. The dataset is synthetically generated in Unreal Engine 5 (Epic Games, 2024), featuring photorealistic rendering with accurate simulation of object geometry, material attributes, lighting, and reflections. The counting criteria for this dataset are similar to those of the ToyShape dataset (see Sec. 4.2.1). Formally, we have $C = \{\text{mug}, \text{apple}, \text{clock}\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in S_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Quantification of counting hallucinations. The definition of counting hallucinations, the construction of the counting model, and the quantification protocol follow the same procedure as in the ToyShape dataset, owing to the similarity of their counting criteria, as shown in Fig. 2.

4.2.3 REALHAND

Dataset description. To probe counting hallucinations in the hardest, real-world regime, we construct RealHand from 5,050 human hand images drawn from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow². Specifically, we aggregate raw images from these sources and remove those with blur or artifacts. Each image depicts a human hand with five fingers, capturing both dorsal (back) and palmar (front) views under varying lighting conditions and backgrounds (e.g., white surfaces, blackboards, walls), as shown in the lower-left of Fig. 1. The counting criteria specify that each image contains exactly five fingers, i.e., $C = \{\text{finger}\}$ with $N_c(\mathbf{x}) \in S_c = 5$ for each $c \in C$, and $\sum_{c \in C} N_c(\mathbf{x}) = 5$.

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

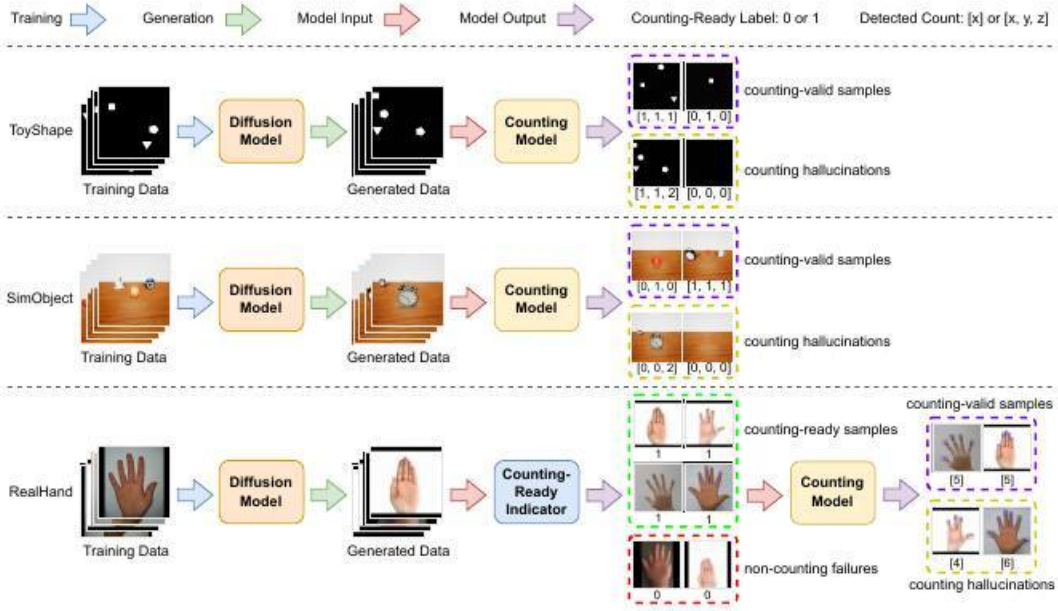


Figure 2: Evaluation procedure in the CountHalluSet suite. For ToyShape and SimObject, generated images are assessed for counting directly, since severe degradations are rare. For RealHand, a counting-ready indicator (a binary classifier aligned with human judgments, > 96% accuracy) first separates images suitable for counting from non-counting failures such as severely deformed fingers, after which the counting model (> 99% accuracy) determines the finger count; this two-stage design lets us report the counting hallucination rate and the non-counting failure rate as two disentangled, individually measurable quantities. Count labels correspond to ToyShape: *triangle, square, pentagon*; SimObject: *mug, apple, clock*; RealHand: *finger*.

Counting hallucinations and non-counting failures. Generating realistic hand images is inherently prone to unexpected failures beyond counting hallucinations (e.g., extra or missing fingers), such as blurriness, severely deformed fingers, and visual artifacts. Therefore, we use a counting-ready indicator (i.e., a binary classifier) to identify these cases. Examples of counting hallucinations and non-counting failure samples can be found in Appendix A.7.

Counting-ready indicator and counting model used in RealHand. To separate counting hallucinations from previously mentioned failure modes, we finetune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images to serve as a counting-ready indicator aligned with human judgments. For counting, we finetune a YOLO-12 (Tian et al., 2025) model on 2k fingertip-annotated generated images. Against human evaluation of 500 randomly sampled images by three annotators, the indicator and the counting model reach over 96% and 99% accuracy, respectively; these high agreement rates confirm that even the most challenging RealHand measurements rest on a reliable labeling pipeline rather than on approximate heuristics. Examples of YOLO detection results can be found in Appendix A.6.

Quantification of counting hallucinations. Unlike the ToyShape and SimObject datasets, where all generated images are directly evaluated due to their simplicity and the rarity of severe failure cases (e.g., distorted or unrecognizable objects), the RealHand dataset requires an additional filtering step. Specifically, we first apply the counting-ready indicator to select hand images with sufficient visual quality, and only then use the counting model to predict the number of fingers, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. We train a DDPM (Ho et al., 2020) from scratch for the ToyShape dataset and fine-tune latent diffusion models (LDMs) (Rombach et al., 2022), pretrained on the

CelebA-HQ dataset (Karras et al., 2018), for the SimObject and RealHand datasets. The time step T is set to 1,000 with a standard linear noise schedule. Images are resized to 128×128 for DDPMs and 256×256 and LDMs. We train the models using the Adam optimizer (Kingma, 2015) with a learning rate of 0.0001, running for 150k, 300k, and 80k steps on the ToyShape, SimObject, and RealHand datasets, respectively, with an effective batch size of 256.

Inference of Diffusion Models. For the ancestral sampling (i.e., DDPM), the sampling steps are set to 1,000. For the ODE-based solvers, we consider both first-order and second-order methods: DPM-Solver-1 (equivalent to DDIM (Song et al., 2021)) as a first-order solver, and DPM-Solver-2 (Lu et al., 2022) as a second-order solver. Each solver is evaluated under three widely-used denoising step configurations, 25, 50, and 100 steps, giving a controlled grid in which solver order and step count vary one at a time. For the RealHand dataset, the counting model employs a YOLO detector with a confidence threshold of 0.3 and an IoU threshold of 0.1. Unless noted otherwise, the number of generated samples matches the number of training samples in each evaluation, and every reported rate is averaged over three random seeds, so the consistent trends we highlight below reflect stable, seed-averaged measurements rather than single runs.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports the counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) across a controlled grid of denoising conditions, spanning two solver orders, three step budgets, and two initial-noise settings on all three datasets (ToyShape, SimObject, RealHand). Reading the table by column exposes several patterns that recur consistently across settings.

(1) More sampling steps lower counting hallucinations on synthetic data but not on RealHand. As shown in Table 1, moving from 25 to 100 steps steadily reduces CHR on the synthetic datasets, from 2.43% down to 1.56% on ToyShape and from 9.27% to 8.04% on SimObject with DPM-Solver-1. On RealHand the same additional computation moves in the opposite direction: CHR rises from 12.71% to 14.29% for DPM-Solver-1 and stays in the 14–16% band for DPM-Solver-2, the only exception being the modest dip at 100 steps with DPM-Solver-2. This divergence indicates that the simpler, more structured synthetic distributions benefit from finer solver granularity, whereas the more complex RealHand distribution does not: extra steps sharpen local detail without correcting finger counts, and can even amplify the error.

(2) Higher-order solvers trade counting fidelity for perceptual quality on RealHand. Holding the step budget fixed and comparing DPM-Solver-1 against DPM-Solver-2 on RealHand exposes a sharp and repeatable trade-off. At 25 steps the higher-order solver cuts NCFR from 16.18% to 8.37% (roughly halving it) while pushing CHR up from 12.71% to 14.23%; at 50 steps NCFR again nearly halves (11.32% → 6.38%) as CHR rises from 13.82% to 16.15%. The same opposing movement appears as steps increase within either solver. In short, spending more computation, whether through a higher solver order or more steps, reliably buys better perceptual quality (lower NCFR) but does *not* buy factual correctness of finger counts.

(3) DDPM dominates every ODE configuration on all three failure metrics. Ancestral sampling (DDPM) records the lowest CHR, NCFR, and TFR in every column of Table 1 without exception: CHR of 0.63% on ToyShape, 4.98% on SimObject, and 10.82% on RealHand, together with a RealHand NCFR of just 2.07% and TFR of 12.88%, well below the best ODE-solver TFR of 22.53% and a fraction of its NCFR (6.38%). Because it is uniformly the strongest setting, DDPM serves as a practical lower bound on the failure rates attainable by these models and marks ancestral sampling as the most effective strategy for minimizing hallucinations.

(4) Ground-truth initial noise consistently lowers all three failure rates. Across Table 1, replacing “normal” standard-Gaussian noise with “diffused” ground-truth initial noise yields lower CHR, NCFR, and TFR in essentially every paired comparison; for instance, DPM-Solver-1 at 25 steps on RealHand drops from a TFR of 31.01% (Normal) to 28.90% (Diffused), with parallel improvements in both CHR and NCFR. Because diffused noise matches the initialization the model actually saw during training, this more informed starting point measurably reduces not only counting hallucinations but non-counting failures as well.

Table 1: Counting hallucination rate (CHR; counting hallucinations as a percentage of all generated samples, lower is better) under a controlled grid of solver configurations and initial-noise settings, reported for all three datasets (ToyShape, SimObject, RealHand). For RealHand we additionally report the non-counting failure rate (NCFR; failure cases flagged by the counting-ready indicator) and the total failure rate (TFR = CHR + NCFR), so the two failure types and their sum can be compared directly. *Diffused* and *Normal* denote the ground-truth initial noise and standard Gaussian noise, respectively, with *Diffused* tending to give the lower failure rate. The bottom row (DDPM) attains the smallest value in every column.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

(5) **Counting hallucinations scale monotonically with the morphological complexity of the target object.** Ranking the datasets by object complexity, the CHR climbs in lockstep: at the DDPM lower bound it rises from 0.63% on the simple ToyShape shapes, to 4.98% on the everyday SimObject items, to 10.82% on RealHand fingers, and the same ordering holds under every ODE-solver setting. This clean, order-preserving gap indicates that the fine-grained detail and intricate spatial relationships of morphologically complex objects make accurate object counts markedly harder for diffusion models to preserve, producing far more frequent additions or omissions.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Counting hallucination is a form of factualness hallucination, so it is natural to ask: *Does a better perceptual metric, such as a lower FID, necessarily imply fewer counting hallucinations by reflecting a smaller distribution gap between generated and training data?* Our measurements answer this cleanly: **no**.

To quantify the relationship, we correlate CHR against perceptual metrics and, side by side, NCFR against the very same metrics, so that the two failure types are held to an identical test. We use two widely adopted perceptual measures, the distribution-based FID (Heusel et al., 2017) and the instance-based MUSIQ (Ke et al., 2021), and, to avoid depending on any single notion of association, three complementary correlation coefficients: Pearson (Lee Rodgers & Nicewander, 1988) (linear), Spearman (Spearman, 1904) (monotonic), and Distance Correlation (Székely et al., 2007) (general nonlinear). The contrast between the two blocks of the correlation table is stark and one-directional.

✚ **CHR is only weakly and inconsistently tied to perceptual metrics.** As the first and third rows of Table 2 show, the CHR–FID Pearson correlation does not even keep a stable sign: it is strongly positive on SimObject (0.87), strongly negative on RealHand (−0.97), and weakly positive on PTI-Hand (0.20), with the Spearman coefficients flipping in the same erratic way (0.84 → −0.97 → −0.14). The Distance Correlation likewise erodes as content grows more complex, from 0.82 (SimObject) to 0.74 (RealHand) to 0.47 (PTI-Hand), and pooling DDPM with the ODE-solver results attenuates every coefficient further. No single, dataset-agnostic relationship between FID and counting hallucinations exists.

✚ **NCFR, by contrast, is strongly and consistently tied to the same metrics.** Where CHR wanders, NCFR tracks perceptual quality almost perfectly and in a single direction across every setting: for NCFR vs. FID, every coefficient is at least 0.88 and reaches up to 0.99 (RealHand Pearson), each with a p-value below 0.01, i.e., high statistical significance. Placed against the unstable CHR correlations directly above them, these uniformly strong values make the dissociation unmistakable, perceptual metrics measure non-counting quality, not counting factuality.

Table 2: Correlation of the counting hallucination rate (CHR) and the non-counting failure rate (NCFR) with perceptual metrics (FID and MUSIQ), reported so the two failure types can be compared under identical tests. Each cell lists the Pearson r , Spearman ρ , and Distance Correlation r_d . The consistent message: CHR correlations are small and change sign across datasets, whereas NCFR correlations are large and stable (mostly ≥ 0.88 in magnitude). Results for PTI-Hand are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to produce 5050 samples, matching the size of the RealHand dataset; FID for PTI-Hand is computed between RealHand images and the model-generated samples. Here p is the p-value for the significance of r or ρ , and $\dagger$ marks DDPM results pooled with those from various ODE solvers across sampling conditions.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject	RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01	-0.99	<0.01	-	-	-	-	-	-
Pearson † (r)	0.49	0.14	-0.53	0.13	-	-	0.96	<0.01	-	-	0.46	0.20	-	-	-0.98	<0.01	-	-	-	-	-	-	-	-
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29	-	-	-	-	-	-
Spearman † (ρ)	0.75	0.01	-0.51	0.15	-	-	0.95	<0.01	-	-	0.40	0.28	-	-	-0.91	<0.01	-	-	-	-	-	-	-	-
Distance Correlation (r_d)	0.89	-	0.97	-	0.47	-	0.99	-	0.91	-	0.93	-	0.42	-	0.97	-	0.99	-	-	-	-	-	-	-
Distance Correlation † (r_d)	0.82	-	0.74	-	-	-	0.95	-	-	-	0.71	-	-	-	0.97	-	-	-	-	-	-	-	-	-

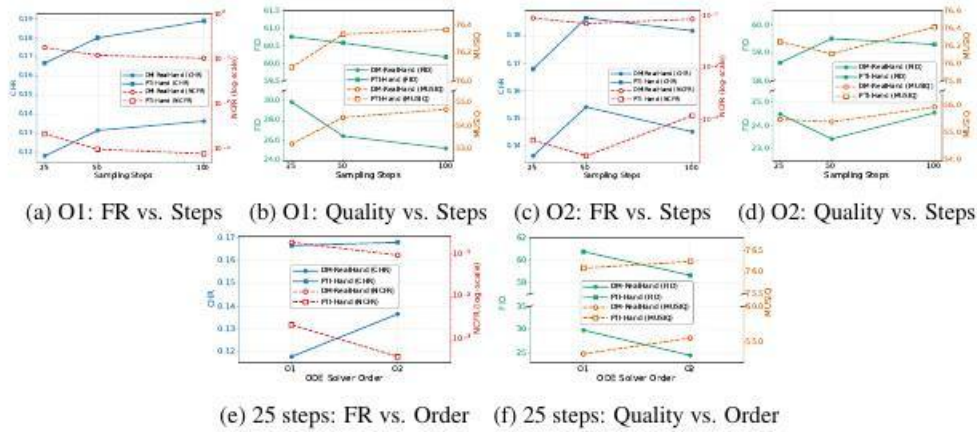


Figure 3: Side-by-side trend comparison between our controlled DM-RealHand and the large pretrained PTI-Hand (Stable-Diffusion-3.5-Medium). O1 and O2 denote first- and second-order ODE solvers (DPM-solver-1 / DPM-solver-2 for DM-RealHand; Euler / Heun for PTI-Hand); FR metrics are CHR and NCFR, quality metrics are FID and MUSIQ. As sampling steps and solver order vary, the two models trace the same curves, i.e., the failure-rate and quality trends move in the same directions in every panel, showing that the patterns quantified on our controlled setup carry over to a large-scale pretrained model.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

A natural question is whether the patterns quantified on our controlled models are artifacts of that setup or genuine properties of diffusion sampling. To test this, we apply the exact counting-ready indicator and counting model from Sec. 4.2.3 to a large-scale pretrained text-to-image model, Stable-Diffusion-3.5-Medium (Esser et al., 2024); more details are in Appendix A.8. The result is a clean replication: as Figure 3 shows, solver type and sampling steps move counting hallucinations in the pretrained model along the same trajectories we measured on controlled RealHand, panel for panel. The correlation numbers in Table 2 reproduce equally well, the PTI-Hand column again shows weak, sign-unstable CHR-metric correlations alongside strong NCFR-metric correlations (e.g., NCFR vs. MUSIQ Distance Correlation of 0.99). This quantitative agreement across an independently trained, orders-of-magnitude larger model confirms that the findings of Sec. 5.2 are not specific to our training regime and that the proposed evaluation protocol transfers reliably to pretrained text-to-image models.

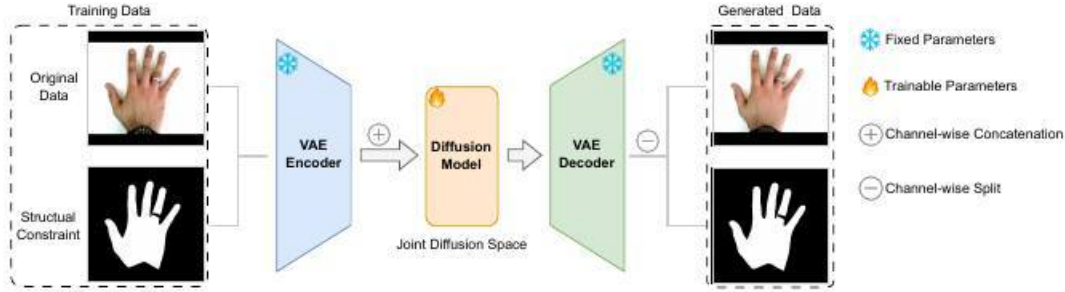


Figure 4: Diagram of the proposed joint-diffusion model (JDM). The model performs channel-wise concatenation of the latent representations of hand RGB images and their corresponding segmentation masks. By learning general appearance and explicit structural constraints jointly, JDM encourages anatomically plausible hand poses and correct finger topology, which translates into the lower CHR and NCFR reported for JDM.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

The monotonic drop in counting hallucination rate toward simpler countable objects, quantified in Observation 5 of Sec. 5.2 (from 10.82% on RealHand down to 0.63% on ToyShape at the DDPM lower bound), suggests that diffusion models capture counting-based factuality far more reliably when the underlying visual structure is morphologically simple (e.g., masked objects). This finding raises an important question: if diffusion models were to denoise general visual representations together with explicit, simple structural constraints that are easier to model, such as object masks that encode basic topology and counts, within a shared latent space, could such a joint learning paradigm mitigate counting hallucinations? We refer to such models that jointly learn both general visual representations and structural constraints during denoising as **joint-diffusion models** (JDM).

Specifically, we perform a channel-wise concatenation of the latent representation of hand mask with that of the corresponding RGB image, as shown in Fig. 4. Then we finetune the UNet pretrained on CelebA-HQ (Karras et al., 2018) (following the RealHand configuration described in Sec. 5.1) using the RealHand images and their associated masks, extracted using SAM-2 (Ravi et al., 2024). To accommodate both modalities, the input and output channels of the denoising UNet are expanded to match the combined latent dimensionality of the image and mask latents (i.e., $3+3=6$ channels).

The gains in Table 3 are large and uniform. Against the vanilla LDM, JDM lowers CHR at every one of the six solver $\times$ step settings (e.g., 14.55% $\rightarrow$ 10.51% for DPM-Solver-1 at 100 steps) and, more dramatically, collapses NCFR from the 7–18% range down to about 2–5%, up to a roughly 6 $\times$ reduction (e.g., 18.06% $\rightarrow$ 2.94% at DPM-Solver-1/25 steps). This across-the-board improvement on both failure metrics demonstrates that jointly modeling structural constraints is highly effective for the structurally intricate objects most prone to counting ambiguities. The single quantity that moves the other way is FID, which rises from the 23–30 range for LDM to 32–38 for JDM. This is an expected cost, not a contradiction of the failure-rate gains: jointly denoising two modalities under a small-scale training regime raises the modeling burden, so the model trades a modest amount of perceptual quality (higher FID) for substantially better structural correctness (far lower CHR and NCFR), precisely the dissociation our correlation analysis predicts. We further observe promising CHR reductions when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning; full details, results, and discussion appear in Appendix A.9.

6 CONCLUSION

In this work, we take the first step toward systematically quantifying hallucinations in diffusion models and offer new, number-driven insights into improving factual generation in this field. Focusing on counting hallucinations, we introduce **CountHalluSet** together with a reproducible measurement protocol whose human-validated counting models ($> 99\%$ accuracy) turn hallucination into an exact, comparable rate. Our controlled experiments deliver a consistent quantitative message: commonly adopted numerical strategies do not mitigate counting hallucinations on human hands (increasing steps from 25 to 100 leaves the RealHand CHR flat or higher, even as it falls steadily on synthetic

Table 3: Head-to-head comparison of the vanilla LDM baseline and our JDM on RealHand across both ODE solvers and all three step budgets (lower is better throughout). JDM wins on CHR and NCFR in every one of the six settings, with especially large NCFR gains (up to $\sim 6\times$), while trading a modest FID increase; teal-bold marks the better method for each metric and column. For JDM, although the model jointly generates hand RGB images and segmentation masks, FID is computed solely on the RGB outputs to keep it directly comparable with the baseline.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

data), and the correlation analysis shows that widely used perceptual metrics such as FID and MUSIQ track non-counting failures strongly (coefficients ≥ 0.88) yet fail to reflect counting hallucinations, whose correlations are weak and even change sign across datasets. This dissociation motivates JDM, which lowers both failure rates on RealHand across every solver and step setting. Experiments on a large-scale pretrained text-to-image model reproduce these trends panel for panel, corroborating both the generality and the validity of our findings.

REFERENCES

- Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J Ballard, Joshua Bambrick, et al. Accurate structure prediction of biomolecular interactions with alphafold 3. *Nature*, pp. 493—500, 2024.
- Mahmoud Afifi. 11k hands: Gender recognition and biometric identification using a large dataset of hand images. *Multimedia Tools and Applications*, 78:20835–20854, 2019.
- Sumukh K Aithal, Pratyush Maini, Zachary C Lipton, and J Zico Kolter. Understanding hallucinations in diffusion models through mode interpolation. *arXiv preprint arXiv:2406.09358*, 2024.
- Ali Borji. Qualitative failures of image generation models and their application in detecting deepfakes. *Image and Vision Computing*, 137:104771, 2023.
- Jake Bruce, Michael D Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, et al. Genie: Generative interactive environments. In *ICML*, 2024.
- Thomas M Cover and Joy A Thomas. *Elements of Information Theory*. John Wiley & Sons, 2012.
- Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *NeurIPS*, 34:8780–8794, 2021.
- Jingtao Ding, Yunke Zhang, Yu Shang, Yuheng Zhang, Zefang Zong, Jie Feng, Yuan Yuan, Hongyuan Su, Nian Li, Nicholas Sukiennik, et al. Understanding world or predicting future? a comprehensive survey of world models. *ACM Computing Surveys*, 2024.
- Epic Games. Unreal engine, 2024. URL <https://www.unrealengine.com>.
- Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *ICML*, pp. 12606–12633, 2024.
- David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2(3), 2018.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pp. 770–778, 2016.
- Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning. In *EMNLP*, pp. 7514–7528, 2021.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *NeurIPS*, 30, 2017.

- Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*, pp. 1–14, 2021.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *NeurIPS*, 33: 6840–6851, 2020.
- Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J Fleet. Video diffusion models. *NeurIPS*, 35:8633–8646, 2022.
- Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural networks*, 2(5):359–366, 1989.
- Kaiyi Huang, Chengqi Duan, Kaiyue Sun, Enze Xie, Zhenguo Li, and Xihui Liu. T2i-compbench++: An enhanced and comprehensive benchmark for compositional text-to-image generation. *IEEE TPAMI*, 2025.
- Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *arXiv preprint arXiv:2311.05232*, 2023.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM computing surveys*, 55(12):1–38, 2023.
- John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with alphafold. *Nature*, 596(7873):583–589, 2021.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- Adam Tauman Kalai and Santosh S Vempala. Calibrated language models must hallucinate. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, pp. 160–171, 2024.
- Tero Karras, Timo Aila, Samuli Laine, and Jaakko Lehtinen. Progressive growing of gans for improved quality, stability, and variation. In *ICLR*, 2018.
- Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. Musiq: Multi-scale image quality transformer. In *ICCV*, pp. 5148–5157, 2021.
- Diederik P Kingma. Adam: A method for stochastic optimization. In *ICLR*, pp. 1–15, 2015.
- Peter E Kloeden, Eckhard Platen, Peter E Kloeden, and Eckhard Platen. *Stochastic differential equations*. Springer, 1992.
- Zhifeng Kong, Wei Ping, Jiaji Huang, Kexin Zhao, and Bryan Catanzaro. Diffwave: A versatile diffusion model for audio synthesis. In *ICLR*, pp. 1–17, 2021.
- Joseph Lee Rodgers and W Alan Nicewander. Thirteen ways to look at the correlation coefficient. *The American Statistician*, 42(1):59–66, 1988.
- Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. Halueval: A large-scale hallucination evaluation benchmark for large language models. In *EMNLP*, pp. 6449–6464, 2023.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *ACL*, pp. 3214–3252, 2022.
- Luping Liu, Yi Ren, Zhijie Lin, and Zhou Zhao. Pseudo numerical methods for diffusion models on manifolds. In *ICLR*, pp. 1–24, 2022.
- Qihao Liu, Adam Kortylewski, Yutong Bai, Song Bai, and Alan Yuille. Discovering failure modes of text-guided diffusion models via adversarial search. In *ICLR*, pp. 1–25, 2024.

- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. Dpm-solver: A fast ode solver for diffusion probabilistic model sampling in around 10 steps. *NeurIPS*, 35:5775–5787, 2022.
- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. Dpm-solver++: Fast solver for guided sampling of diffusion probabilistic models. In *ICLR*, pp. 1–19, 2023.
- Shitong Luo and Wei Hu. Diffusion probabilistic models for 3d point cloud generation. In *CVPR*, pp. 2837–2845, 2021.
- Supreeth Narasimhaswamy, Uttaran Bhattacharya, Xiang Chen, Ishita Dasgupta, Saayan Mitra, and Minh Hoai. Handdiffuser: Text-to-image generation with realistic hand appearances. In *CVPR*, pp. 2468–2479, 2024.
- Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *ICML*, pp. 8162–8171, 2021.
- Michael A Nielsen. *Neural networks and deep learning*, volume 25. Determination press San Francisco, CA, USA, 2015.
- Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *CVPR*, pp. 10684–10695, 2022.
- Mehdi SM Sajjadi, Olivier Bachem, Mario Lucic, Olivier Bousquet, and Sylvain Gelly. Assessing generative models via precision and recall. *NeurIPS*, 31, 2018.
- Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training gans. *NeurIPS*, 29, 2016.
- Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *ICML*, pp. 2256–2265, 2015.
- Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *ICLR*, pp. 1–22, 2021.
- Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *ICLR*, pp. 1–36, 2022.
- C Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904.
- Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *CVPR*, pp. 2818–2826, 2016.
- Gábor J Székely, Maria L Rizzo, and Nail K Bakirov. Measuring and testing dependence by correlation of distances. *The Annals of Statistics*, pp. 2769–2794, 2007.
- Yunjie Tian, Qixiang Ye, and David Doermann. Yolov12: Attention-centric real-time object detectors. *arXiv preprint arXiv:2502.12524*, 2025.
- Zhengzhong Tu, Hossein Talebi, Han Zhang, Feng Yang, Peyman Milanfar, Alan Bovik, and Yinxiao Li. Maxvit: Multi-axis vision transformer. In *ECCV*, pp. 459–479, 2022.
- Arash Vahdat, Francis Williams, Zan Gojcic, Or Litany, Sanja Fidler, Karsten Kreis, et al. Lion: Latent point diffusion models for 3d shape generation. *NeurIPS*, 35:10021–10039, 2022.
- Shirley Wu, Mert Yuksekgonul, Linjun Zhang, and James Zou. Discover and cure: Concept-aware mitigation of spurious correlation. In *ICML*, pp. 37765–37786, 2023.

-
- Dmitry Yarotsky. Universal approximations of invariant maps by neural networks. *Constructive Approximation*, 55(1):407–474, 2022.
- Arman Zarei, Keivan Rezaei, Samyadeep Basu, Mehrdad Saberi, Mazda Moayeri, Priyatham Kattakinda, and Soheil Feizi. Understanding and mitigating compositional issues in text-to-image generative models. *arXiv preprint arXiv:2406.07844*, 2024.
- Qinsheng Zhang and Yongxin Chen. Fast sampling of diffusion models with exponential integrator. In *ICLR*, pp. 1–33, 2023.

A APPENDIX

A.1 POOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions of the ideal reverse process $q(\mathbf{x}_{T:0})$ and the true reverse process $p_\theta(\mathbf{x}_{T:0})$, their KL divergence is defined as:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

For both reverse processes $q(\mathbf{x}_{T:0})$ and $p_\theta(\mathbf{x}_{T:0})$, we can decompose them as follows, leveraging the Markov property:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

By applying the chain rule to decompose the KL divergence, we have:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

Then, according to the Data Processing Inequality (Cover & Thomas, 2012), marginalizing (or discarding some random variables) will not increase the KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

The bound of total accumulated distributional error can be obtained by combining Eq. (A.4) and Eq. (A.5):

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

where equality (=) would strictly hold only in the ideal case where $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for all t and $\mathbf{x}_t$, which means the diffusion model can accurately recover the underlying data distribution $q(\mathbf{x}_0)$. This upper bound for the overall KL divergence ($\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0))$) between true data and model-generated data is identical to the variational bound derived in DDPM (Ho et al., 2020) when data entropy $H(\mathbf{x}_0)$ is excluded, thereby suggesting that optimizing DDPM’s variational loss directly tightens this explicit upper bound on the discrepancy between trued data and model-generated data.

Diffused-prior gap. The first source of errors arises from the mismatch between the distribution of the ground-truth diffused data $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ and the prior distribution used for sampling $p_\theta(\mathbf{x}_T)$. This mismatch stems from the fact that the number of diffusion steps T is finite in practice. The discrepancy can be formally quantified using a general distributional metric:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes the Kullback–Leibler (KL) divergence. Alternatively, other appropriate metrics such as FID or the Wasserstein distance can be employed to evaluate this distributional discrepancy. The diffused-prior gap, present from the outset, induces a deviation that persists throughout the denoising process, ultimately resulting in a sustained output error.

Noise model prediction error. The second source of errors stems from the noise prediction model. Because the model’s estimate $\epsilon_\theta(\mathbf{x}_t, t)$ only approximates the true noise ϵ , as guaranteed in principle by the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022). Let $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$ represent this prediction error. Based on Eq. (5), any discrepancy in ϵ_θ induces a corresponding deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\alpha_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Since each reverse step uses the mean from the previous iteration, this bias accumulates throughout the denoising trajectory, ultimately contributing to the overall prediction error.

A.2 POOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Based on Eq. (7), we take a one-step numerical solver approximates the solution as example. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h^*(\mathbf{x}(\tau), \tau)$ and starts from a ground-truth initial condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Similarly, $\hat{\mathbf{x}}_k$ represent the numerical trajectory computed by a numerical ODE solver at discrete time steps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$ and $\hat{h}$ is the estimated model function. For a specific numerical trajectory from $\hat{\mathbf{x}}_0 - \hat{\mathbf{x}}_N$, we want to calculate its total accumulated trajectory error. Let $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ be the trajectory error at step k . Consider the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. Using the variation-of-constants formula for the exact solution over one step (from τ_k down to τ_{k+1}):

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$. Note that the integration limits are backward in time.

Numerical path evolution. A one-step numerical solver approximates the solution. This numerical step deviates from the exact evolution governed by $\hat{h}$ starting from $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h})$ be the exact solution at τ_{k+1} of the ODE using $\hat{h}$, starting from $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method introduces a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

So, we have:

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtract the ideal evolution Eq. (A.9) from the numerical evolution Eq. (A.12):

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. We can unroll this recurrence relation from $k = 0$ to $N - 1$. Let $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$, and $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$. Applying the recurrence

repeatedly:

$$e_N = \mathcal{G}(\tau_N, \tau_0)e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated initial error term represents the deterministic impact on the final path error of a specific initial error instance, whose statistical origin lies in the distributional Diffused-Prior Gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As shown in Eq. (2), the forward process in DDPM (Ho et al., 2020) can be formulated as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

And the reverse process is defined as follows:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\bar{\alpha}_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to that in Eq. (A.16), while its deterministic reverse process (with $\eta = 0$) is defined as follows:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \epsilon_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share the same forward (diffusion) process, DDIM does not simply strip the stochastic term from DDPM's reverse (denoising) process. This is because the two methods employ different coefficient formulations for the predicted noise term.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 presents representative examples from the three datasets used in this study. The category counts in Table A1 are near-uniform: for ToyShape and SimObject every category lands within roughly 1% of an even split (e.g., 20,027 / 19,902 / 19,978 for ToyShape), so no shape or object dominates the training signal. Table A2 shows the same balance by object count, with images containing 1, 2, or 3 objects each near 10,000, confirming that our reported hallucination rates are not confounded by an underlying object-count bias in the training data. The datasets appear in the same order in both tables, and the category order in Table A1 is defined as follows:

- **ToyShape:** triangle, square, pentagon;
- **SimObject:** mug, apple, clock;
- **RealHand:** finger.

Table A1: Per-category object counts for each dataset, showing a near-uniform split (within $\sim 1\%$ of even for ToyShape and SimObject).

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Number of images grouped by object count per dataset, again roughly balanced ($\approx 10,000$ per count for ToyShape and SimObject).

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

We provide representative examples of the counting hallucinations counted in our rates, one panel per dataset, in Fig. A2. On ToyShape and SimObject, a hallucination is any image with two or more instances of some category or with no object at all despite a normal background, as shown in Fig. A2(a) and Fig. A2(b). On RealHand, hallucinations are precisely the images whose finger count departs from five (e.g., 4 or 6 fingers), as shown in Fig. A2(c). Notably, the last two images of the first row in Fig. A2(c) appear to display five fingers; however, one finger is partially incomplete and visibly artifacted, effectively resulting in four valid fingers. Consequently, these samples are classified as counting hallucinations.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative examples of YOLO predictions on the RealHand dataset, which visually substantiate the $> 99\%$ counting accuracy reported in Sec. 4.2.3. The detector localizes fingertips correctly across diverse hand poses and orientations, spanning both dorsal and palmar views. The residual errors are confined to genuinely ambiguous anatomy, such as hands with duplicated thumbs (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)) or hands with five middle fingers but no thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)); these still contain five fingertips and are therefore correctly counted as five-finger hands.

Figure A3(b) shows counting-hallucinated samples in which YOLO returns the correct incorrect count (e.g., 3, 4, 6, or 7 fingertips), i.e., it flags exactly the deviations that define a counting hallucination. Figure A3(c) illustrates the small tail of failure cases, fewer than 1%, where unclear boundaries, cracked fingers, or floating artifacts defeat the detector. Tuning YOLO’s inference hyperparameters (e.g., IoU and confidence thresholds) can recover some of this $< 1\%$ tail, but at the cost of accuracy on the overwhelming majority of counting-valid samples. Because a single fixed detection setup is applied uniformly across every RealHand evaluation, this sub-1% residual is identical across all compared conditions and therefore cannot alter the relative rates or the conclusions drawn from them.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 shows the three mutually exclusive classes that our two-stage RealHand protocol assigns every generated image to: counting-valid, counting-hallucinated, and non-counting failure. This partition is what lets us report CHR and NCFR as separate, non-overlapping quantities. As illustrated in Fig. 2, the counting-ready indicator first splits counting-ready samples (counting-valid plus counting-hallucinated) from images unsuitable for counting. The latter, which populate the NCFR,

typically show severe visual artifacts such as distorted, cracked, partially visible, or floating fingers. Counting-ready samples are then labeled counting-valid or counting-hallucinated purely by whether the detected number of valid fingertips equals five, so the CHR isolates count errors from quality failures.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

In the PTI-Hand experiments, we adopt 10 different hand-generation prompts to prompt Stable-Diffusion-3.5-Medium. The prompts are show as follows.

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

The definition of hallucinations assumes that the model generates patterns that were never observed during training. However, this assumption is difficult to verify for large-scale pretrained diffusion models, where the full training distribution is inaccessible. To keep the PTI-Hand counting hallucination rate a conservative, defensible measurement, we count only the unambiguous cases: a generated hand is labeled a counting hallucination solely when it shows six or more fingers. This one-sided criterion is deliberately strict, real images can show fewer than five visible fingers because of occlusion, articulation, or viewpoint, so under-counts are excluded to avoid inflating the rate, whereas six-plus fingers essentially never occur in real hands. Finger counts ≥ 6 therefore give an empirically verifiable lower-bound signal of counting hallucination, meaning the reported PTI-Hand rates, if anything, understate the true prevalence. Generated examples can be found in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

In the JDM experiments reported in Table 3, we fine-tune the denoising UNet for 120k steps, $1.5\times$ the 80k steps used for the vanilla LDM. This larger training budget is required because doubling the latent space from three to six channels introduces a substantial mismatch with the pretrained latent distribution of LDM. The additional optimization closes this distributional gap so that JDM’s modeling capacity fully materializes; the large CHR and NCFR reductions reported in the JDM comparison are measured after this convergence.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

Importantly, JDM should be read as a **novel training paradigm** for improving factual generation in diffusion models, not as a heavier inference strategy, so its CHR and NCFR reductions come essentially for free at test time. On the one hand, JDM extends only the input and output channels of the UNet while leaving every intermediate layer size unchanged, so it **introduces negligible**

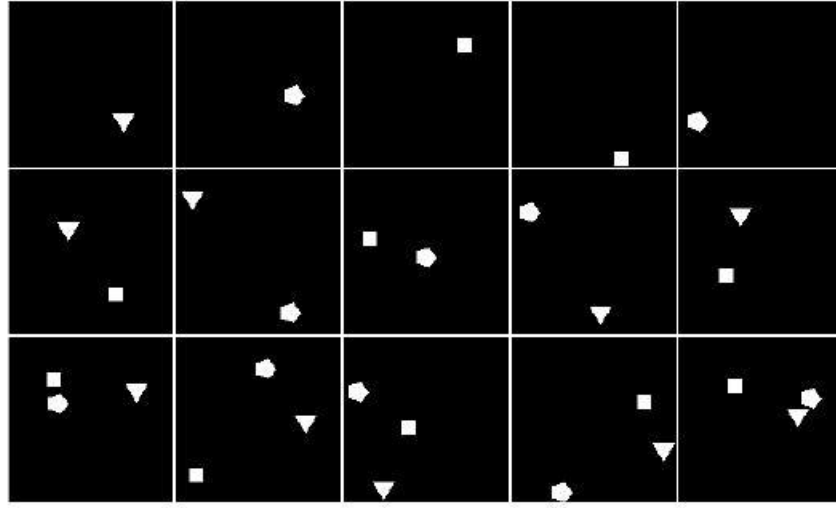
Table A3: Failure rates and image quality for SD-1.5 versus SD-1.5 with joint-diffusion finetuning on RealHand (lower is better). The first row (baseline) finetunes Stable-Diffusion-1.5 on RealHand; the second freezes all intermediate layers of that baseline and finetunes only the input/output layers with extended channels. Even under this heavily constrained recipe, joint-diffusion finetuning lowers CHR at 25 and 50 steps for both solvers (e.g., 0.3362 $\rightarrow$ 0.2838 at DPM-Solver-1/25 steps), while, as expected, NCFR and FID rise, mirroring the perceptual-vs-factual trade-off seen in the main JDM comparison on RealHand.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fr�chet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

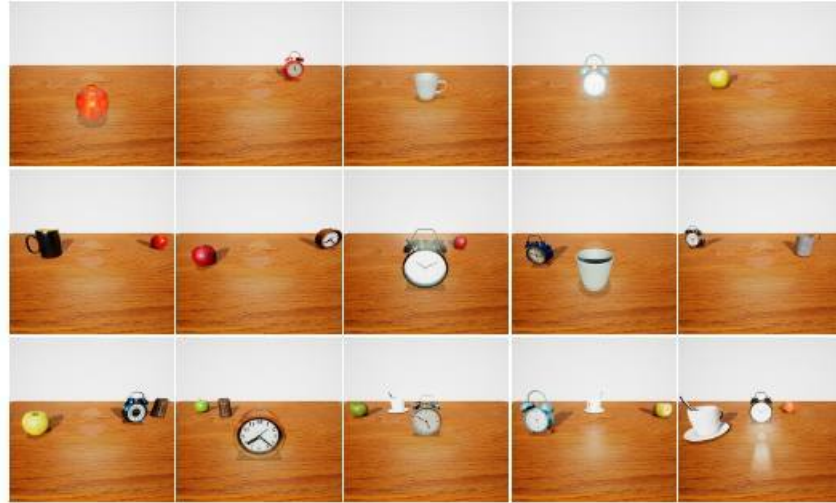
inference-time overhead relative to vanilla LDM even as it substantially lowers both failure rates. On the other hand, adding mask channels to a pretrained text-to-image model **disrupts the manifold geometry of the pretrained latent space and undermines the original denoising distribution**, which makes direct finetuning nontrivial and explains the constrained recipe used below.

Despite this challenge, JDM still delivers measurable counting hallucination reductions on large-scale pretrained diffusion models under carefully controlled finetuning. The difficulty is that aligning JDM’s extended latent space with the rigid latent geometry of a large pretrained model is hard at the limited scale of RealHand. We therefore adopt a two-stage strategy: first finetune Stable-Diffusion-1.5 on RealHand, then freeze all intermediate UNet layers and finetune only the input/output layers with extended channels. As Table A3 shows, even this deliberately constrained recipe lowers CHR at 25 and 50 sampling steps for both solvers, confirming that the counting-fidelity benefit of joint diffusion survives the transfer to a large pretrained backbone.

Taken together with the RealHand JDM results, this evidence positions JDM as a promising, empirically supported direction for enforcing factual consistency in image generation. Because these gains are obtained under a small-scale, heavily constrained regime, we regard **quantifying JDM’s behavior under true large-scale training as an important and compelling avenue for future research**.



(a) Examples in the ToyShape dataset

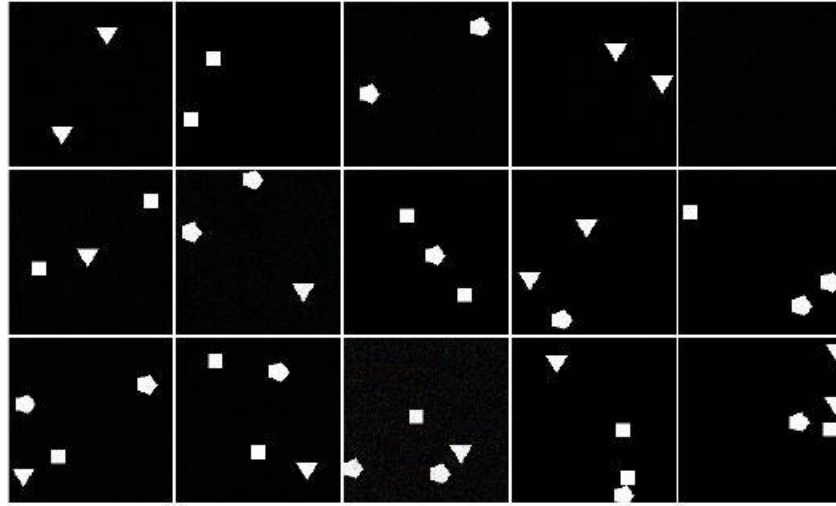


(b) Examples in the SimObject dataset



(c) Examples in the RealHand dataset

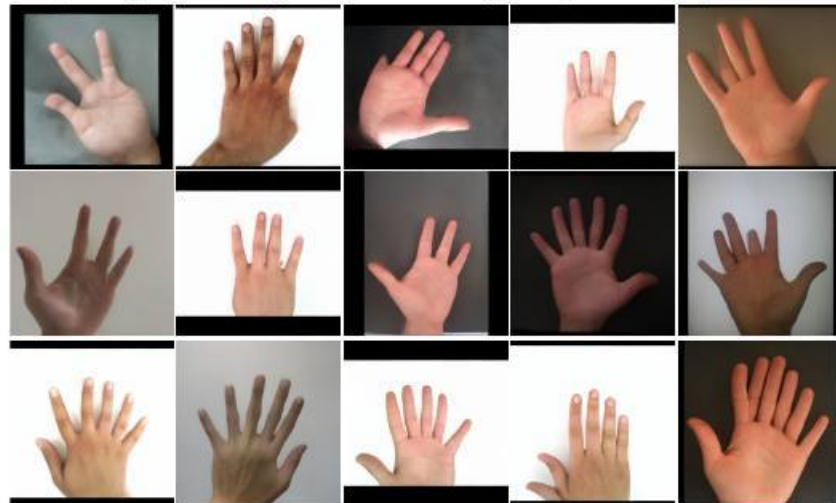
Figure A1: Examples of datasets in the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) Examples of counting hallucinations corresponding to the ToyShape dataset

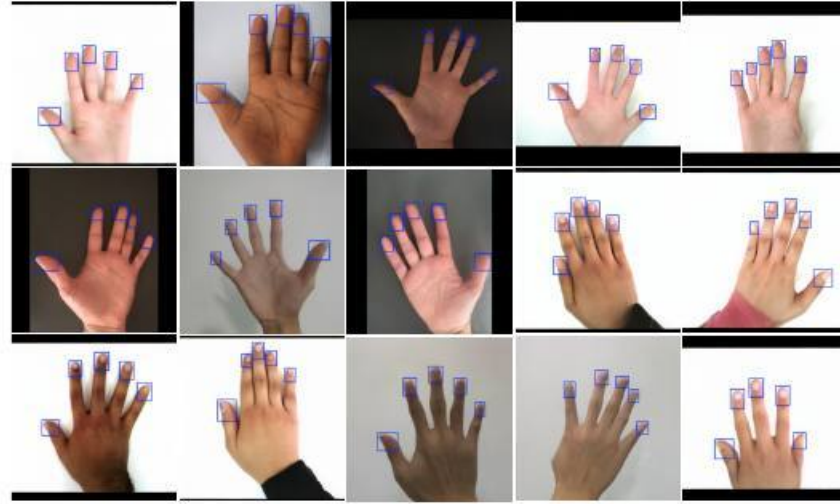


(b) Examples of counting hallucinations corresponding to the SimObject dataset

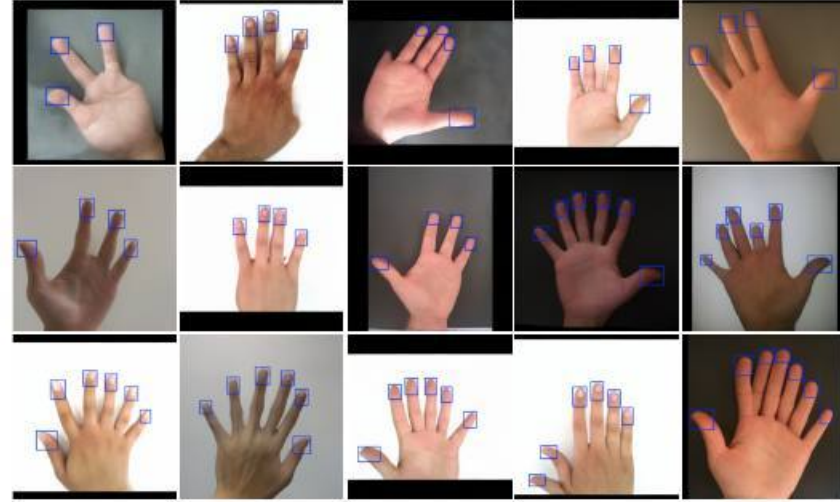


(c) Examples of counting hallucinations corresponding to the RealHand dataset

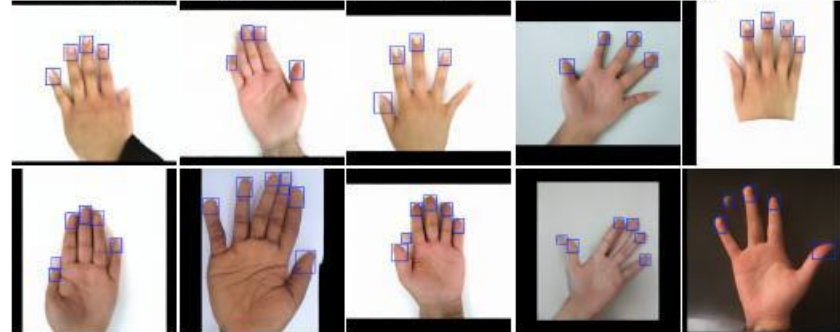
Figure A2: Examples of counting hallucinations corresponding to the ToyShape, SimObject, and RealHand datasets.



(a) Counting-valid YOLO predictions with five fingers

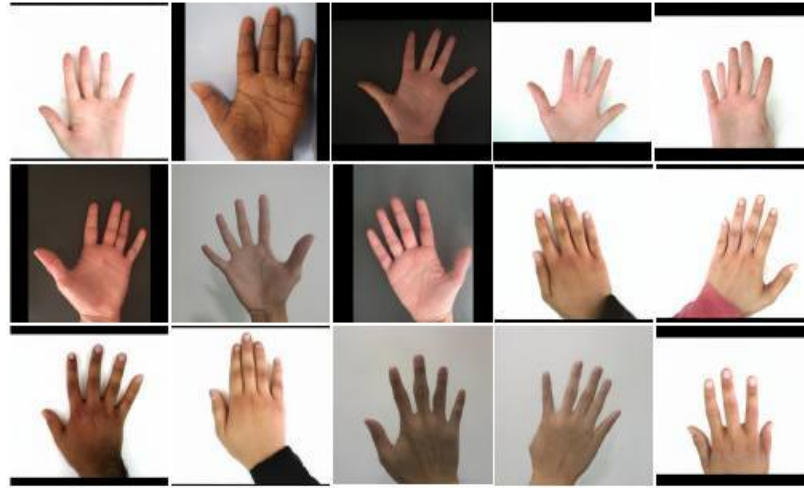


(b) Counting-hallucinated YOLO predictions with not five fingers

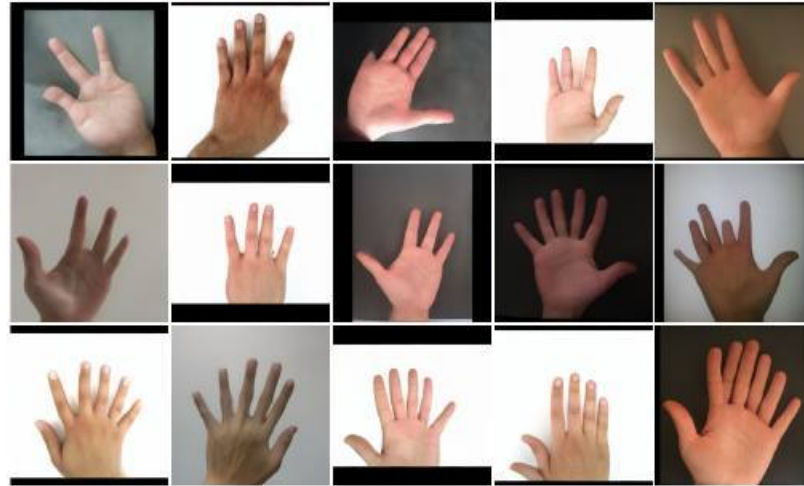


(c) Rare edge cases where YOLO predictions fail

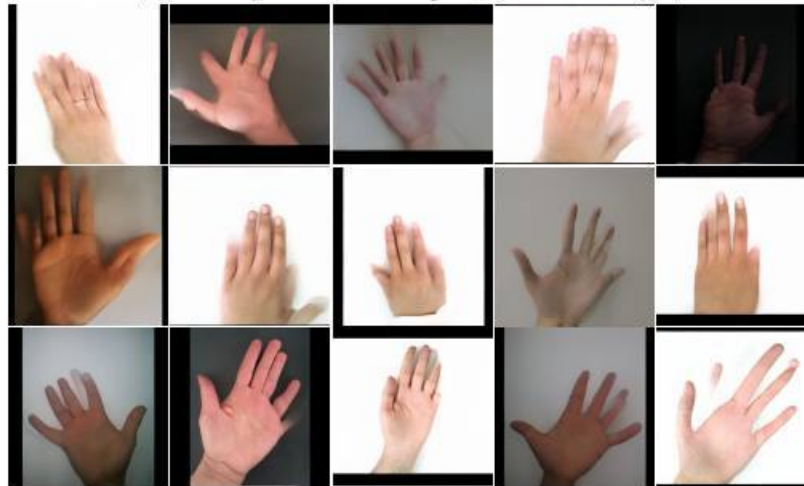
Figure A3: Examples of YOLO predictions on generated RealHand images. (a) Counting-valid samples with five fingertip bounding boxes correctly predicted by YOLO (including cases with two thumbs). (b) Counting-hallucinated samples with incorrect fingertip count predictions (e.g., 3, 4, 6, or 7 detected fingertips). (c) Several rare edge cases where the counting-ready indicator fails to filter out and YOLO struggles to handle, such as blurred, incomplete, or anatomically disconnected fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (not five valid fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A4: Examples of three classes of generated hand images. (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with an incorrect number of valid fingers (e.g., 4 or 6). (c) Non-counting failures: images unsuitable for counting due to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (more than 5 fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A5: Examples of three classes of generated hand images using pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with more than five valid fingers (e.g., 6 or 7). (c) Non-counting failures: images unsuitable for counting due to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.